

Interpretable Shipment ETA and Delay Propagation Prediction

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Abstract. This report presents a unified Extra Trees pipeline for three logistics tasks: predicting final shipment duration and ETA, estimating proxy delay and severe-delay risk, and updating downstream delay from checkpoint observations. Checkpoint records are aggregated into 14,817 trips, evaluated with a chronological split, and protected from future-only features. On the final test set, ETA reaches **MAE 104.93 minutes and R^2 0.898**, reducing MAE by **72.7%** versus the OSRM baseline. Severe-delay classification achieves **F1 0.899 and ROC-AUC 0.911**. The remaining-delay propagation model improves from **MAE 100.80 / R^2 0.800** near the 25% target to **MAE 85.95 / R^2 0.852** near trip completion.

1. Problem Definition

Task	Definition and output
ETA	Duration = TripEndTime - TripStartTime; ETA = TripStartTime + predicted duration. Metrics: MAE, MedianAE, RMSE, R^2 .
Delay magnitude	ProxyDelay = ActualDuration - OSRMTime. Predicted delay is derived from the ETA model, preserving algebraic consistency.
Severe delay	Label = 1 when ProxyDelay > 120 minutes. Metrics: precision, recall, F1, ROC-AUC, and PR-AUC.
Propagation	At checkpoint k, predict the remaining delay; final delay = observed cumulative stage delay + predicted remaining delay.

Assumption. OSRM is a routing baseline, not an official delivery SLA. The 120-minute threshold is an adjustable business proxy.

2. Dataset and Unit of Analysis

144,867 checkpoint rows	26,369 aggregated legs	14,817 unique trips	24 raw columns
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The data flow is **checkpoint rows -> legs -> trips**. Each trip_uuid becomes one row before splitting. Ground-truth duration is calculated directly from start and end timestamps. The aggregated actual_time field is retained only for auditing because its median is about 117 minutes below full elapsed duration.

3. Data Quality and Preprocessing

- Mixed-format timestamps are parsed without missing values; checkpoints are ordered by increasing actual_time within each leg.
- No exact duplicate rows are found. Missing source and destination names are filled from center codes before extracting city and state.
- Numeric inputs use median imputation and standardization. Categorical inputs use most-frequent imputation and one-hot encoding with unseen-category handling.
- For propagation, 21 negative actual-time increments and 2,347 non-positive planned-time rows are flagged. After all stage rules, 2,365 rows are removed, leaving 142,502 stages.
- Negative proxy delays are kept, so early arrivals are not clipped to zero.

4. Start-of-Trip Features and Leakage Guard

Group	Features
Plan	osrm_time, osrm_distance, planned_speed_kmph, planned_minutes_per_km
Time	start_hour, day of week, month, weekend flag
Route	route_type, source_state, destination_state, state_corridor

Leakage guard. Start-of-trip models exclude actual_time, segment_actual_time, trip_end_time, delay_minutes, actual ETA, leg count, and checkpoint count. Actual checkpoint information is used only after it has been observed.

5. Exploratory Data Analysis

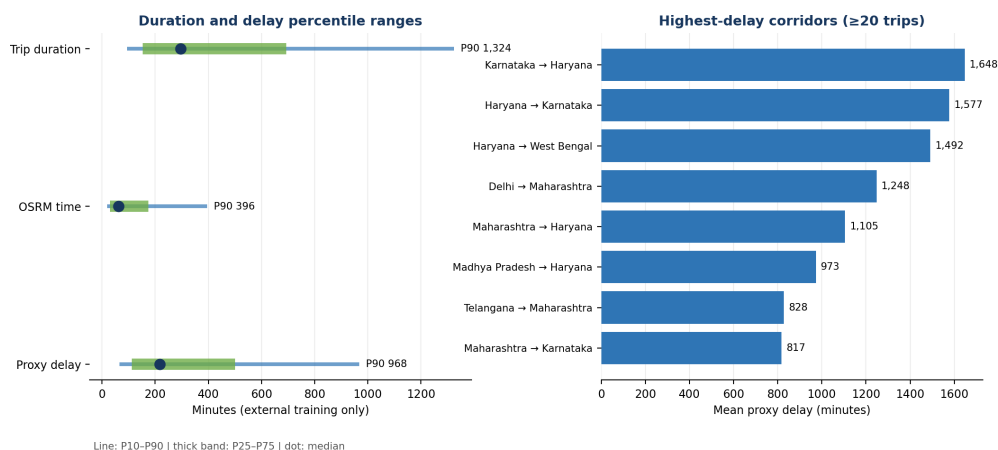


Figure 1. EDA uses only the 10,654 external-training trips. Median duration is 294.90 minutes, median OSRM time is 62.00 minutes, and median proxy delay is 217.65 minutes. The wide interquartile and P90 ranges, together with high-delay corridors, show that route context is essential beyond the OSRM estimate.

6. Chronological Evaluation Protocol

Split	Trips	Date range	Purpose
Fit	8,523	Sep 12-23	Train the candidate
Validation	2,131	Sep 23-26	Select threshold and verify configuration
Final test	4,163	Sep 27-Oct 3	Evaluate once

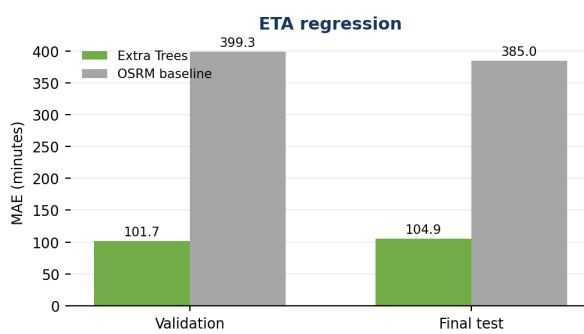
There is no temporal shuffling and no overlap in trip_uid. The final test set is not used for threshold selection. Twelve start-of-trip input columns expand to 157 sparse features when the preprocessor is refit on all external-training trips.

8. Experimental Results

7. Focused Extra Trees Design

Task	Model / reference
ETA	Extra Trees Regressor / OSRM baseline
Severe delay	Extra Trees Classifier, balanced training weights
Propagation	Extra Trees Regressor for remaining delay / current cumulative-delay baseline

Scope. The evaluation reports Extra Trees against the OSRM and cumulative-delay baselines under the same chronological protocol. Performance claims are limited to these tested comparisons.



Final-test confusion matrix

Actual \ Predicted	Not severe	Severe
Not severe	796	385
Severe	230	2,752

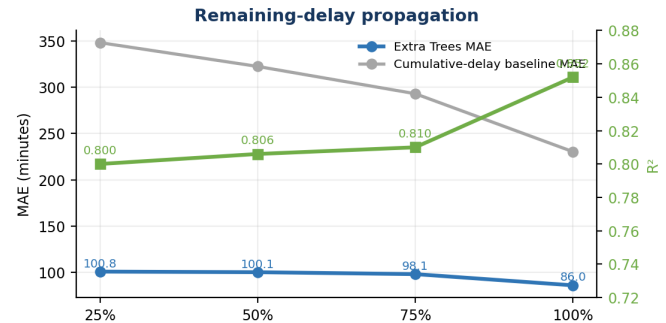
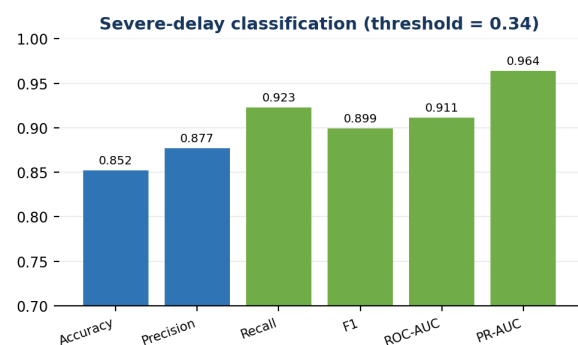


Figure 2. Validation and final-test ETA errors are close. The severe-delay classifier prioritizes recall, while the remaining-delay propagation model consistently outperforms the cumulative-delay baseline and becomes more accurate as additional trip progress is observed.

ETA and Delay Magnitude

Task / model	MAE	MedianAE	RMSE	R ²
ETA - Extra Trees, validation	101.74	52.72	183.09	0.926
ETA - Extra Trees, final test	104.93	53.04	223.81	0.898
ETA - OSRM baseline, final test	385.00	208.91	599.94	0.266
Delay magnitude - derived prediction	104.93	53.04	223.81	0.763

Main improvement: Extra Trees reduces final-test ETA MAE by 72.7% relative to OSRM. Because delay is calculated from the same duration prediction, ETA and delay outputs cannot contradict each other algebraically.

Severe Delay and Propagation

Severe delay: threshold 0.34; Accuracy 0.852; Precision 0.877; Recall 0.923; F1 0.899; ROC-AUC 0.911; PR-AUC 0.964. The final test contains 2,752 true positives, 230 false negatives, 796 true negatives, and 385 false positives.

Target progress	Mean selected	MAE	R ²
25%	38.3%	100.80	0.800
50%	49.8%	100.13	0.806
75%	65.4%	98.07	0.810
100%	98.9%	85.95	0.852

Each trip contributes one nearest available checkpoint at each target. Sparse checkpoint schedules explain why the mean selected progress can differ from the nominal target.

9. Global Explainability

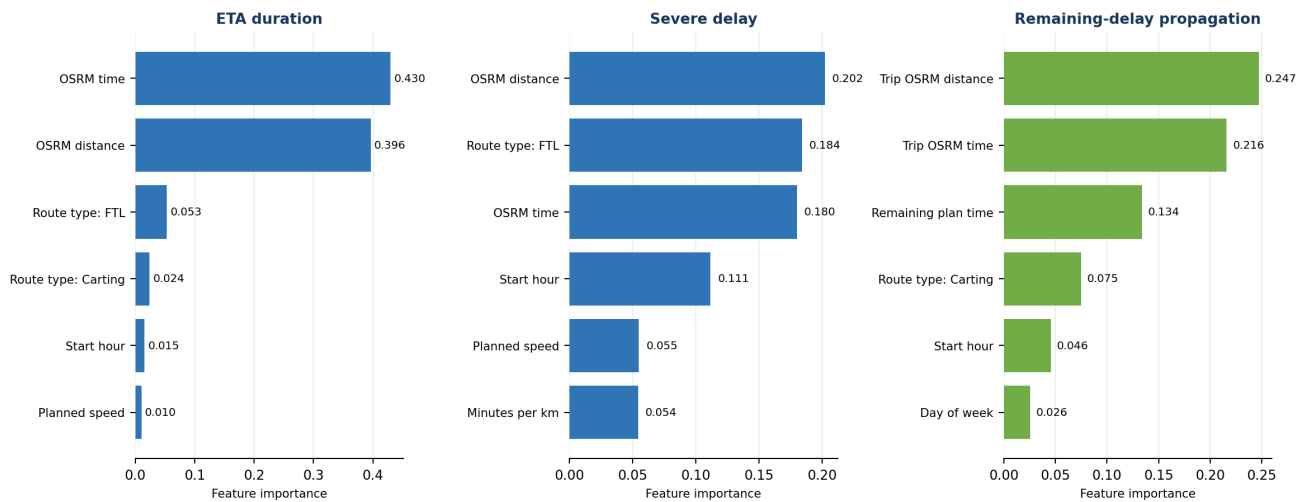


Figure 3. ETA is dominated by OSRM time and distance. Severe-delay risk also depends strongly on route type and departure time. For propagation, total route scale and remaining planned time are more important than any single observed checkpoint feature.

10. Local Operational Example

Field	Value
Route	Pappadahandi, Orissa -> Visakhapatnam, Andhra Pradesh
OSRM baseline	211.00 minutes
Predicted duration	804.65 minutes
Predicted proxy delay	593.65 minutes
Actual proxy delay	7,687.55 minutes

Local SHAP attributes the predicted duration mainly to OSRM distance (+148.81 minutes) and OSRM time (+119.04 minutes), followed by departure hour, route type, and planned-speed ratios. The very large residual error shows that extreme disruptions cannot be explained fully without external operational signals.

Actionable interpretation. The system separates baseline route difficulty from delay accumulated during execution. Dispatch teams can update ETA after each checkpoint and prioritize trips whose predicted remaining delay stays high despite progress.

11. Correctness and Reproducibility

Trip integrity

One row per trip for ETA; one split per trip.

Leakage assertions

Targets and future-only features are excluded.

Balanced contribution

Stage weights prevent long trips from dominating.

Time order

Fit precedes validation; validation precedes final test.

Stage logic

Progress is bounded; invalid stage increments are removed.

Reproducibility

Fixed random state, saved preprocessors, models, metrics, and metadata.

12. Implementation Summary

- Checkpoint propagation uses an **Extra Trees Regressor** to estimate **remaining delay**; final delay combines observed cumulative stage delay with this estimate.
- The severe-delay classifier applies a validation-selected probability threshold of **0.34**.
- Final-test ETA performance is MAE 104.93, MedianAE 53.04, RMSE 223.81, and R^2 0.898.
- At the checkpoint nearest the 100% target, propagation reaches MAE 85.95 and R^2 0.852.
- EDA and model decisions use external-training data, while the final test set is reserved for one-time evaluation.

13. Limitations and Future Work

- OSRM is only a routing proxy; the severe-delay threshold must be calibrated to the real service policy.
- The September-October 2018 dataset lacks weather, traffic, congestion, customs, load, and vehicle-condition variables.
- Extreme outliers remain difficult; robust objectives, anomaly-specific models, or prediction intervals should be evaluated.
- The propagation progress feature assumes the route plan is known and checkpoint coverage is sufficiently dense.
- Useful extensions include probability calibration, conformal prediction, drift monitoring, weather/traffic APIs, and an online checkpoint ETA endpoint.

14. Conclusion

The proposed system addresses final ETA, proxy delay magnitude, severe-delay risk, and checkpoint-based delay propagation in one consistent pipeline. Its reliability comes from correct timestamp-based ground truth, trip-level aggregation, chronological evaluation, explicit leakage controls, sparse preprocessing, and interpretable feature analysis. Under this protocol, **Extra Trees substantially outperforms the OSRM and cumulative-delay baselines** while providing operationally useful explanations of route scale, timing, and remaining delay.

Scientific claim boundary. The report does not claim superiority over published systems because no matched comparison with the same dataset, target, and split protocol has been established.